JAMES EDWARD RICHARDSON

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PROFESSIONAL SUMMARY

I'm a machine learning engineer with 8 years of experience building production systems and leading technical initiatives across fintech and enterprise sectors. I got into this field because I was fascinated by how algorithms could solve real business problems, and I've spent the last decade proving that fascination right. Outside of work, I'm into rock climbing and spend most weekends at local climbing walls or planning trips to Wales. I'm looking for roles where I can mentor junior engineers while staying hands-on with challenging ML problems.

WORK EXPERIENCE

Principal Machine Learning Engineer

DataFlow Systems Ltd

www.dataflowsystems.io

London, United Kingdom

June 2023 - Present

- Led technical architecture and design decisions for machine learning platform serving 150+ enterprise clients with 500+ deployers across the organisation

- Managed and mentored a team of 25 machine learning engineers, data scientists, and ML infrastructure specialists across three geographic locations

- Implemented end-to-end MLOps pipeline reducing model deployment time from 6 weeks to 48 hours using Kubernetes and Apache Airflow

- Designed and deployed real-time fraud detection system processing 2 million transactions daily with 99.97% uptime SLA

Machine Learning Engineer

Insight Analytics Group

www.insightanalytics.net

Oxford, United Kingdom

January 2021 - May 2023

- Developed and deployed natural language processing models for customer sentiment analysis serving 45 million API requests monthly

- Conducted exploratory data analysis and feature engineering on datasets containing 2.8 billion customer records using PySpark

- Optimized XGBoost classification model achieving 94.2% precision and 91.8% recall for credit risk assessment

- Built Python package for model evaluation metrics and distributed it internally across 8 teams using GitLab CI/CD

Junior Data Science Associate

TechVision Consulting

www.techvisionconsulting.co

Manchester, United Kingdom

August 2018 - December 2020

- Collaborated with senior analysts to develop predictive models for customer churn using logistic regression and random forests

- Cleaned and preprocessed datasets using Pandas and NumPy, improving data quality by 34%

- Created interactive dashboards in Tableau to visualize model performance metrics for stakeholder presentations

- Assisted in deployment of two machine learning models to production environments using Docker containerization

EDUCATION

MSc Machine Learning and Computational Statistics

University of Edinburgh

Edinburgh, Scotland

Graduated: September 2017

BSc Computer Science

University of Manchester

Manchester, England

Graduated: June 2015

TECHNICAL SKILLS

Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM, Random Forests, Neural Networks, Gradient Boosting, Clustering Algorithms

Data Processing: Python, PySpark, Pandas, NumPy, SQL, Apache Spark, Hadoop

Deep Learning: Convolutional Neural Networks, Recurrent Neural Networks, Transformers, BERT, Transfer Learning

MLOps and Deployment: Kubernetes, Docker, Apache Airflow, MLflow, Jenkins, GitLab CI/CD, AWS SageMaker, Google Cloud AI Platform

Data Visualization: Tableau, Matplotlib, Seaborn, Plotly

Big Data Technologies: Spark SQL, Hive, HBase, Kafka

Additional Skills: Statistics, A/B Testing, Experimental Design, Model Evaluation, Feature Engineering